

Final Review

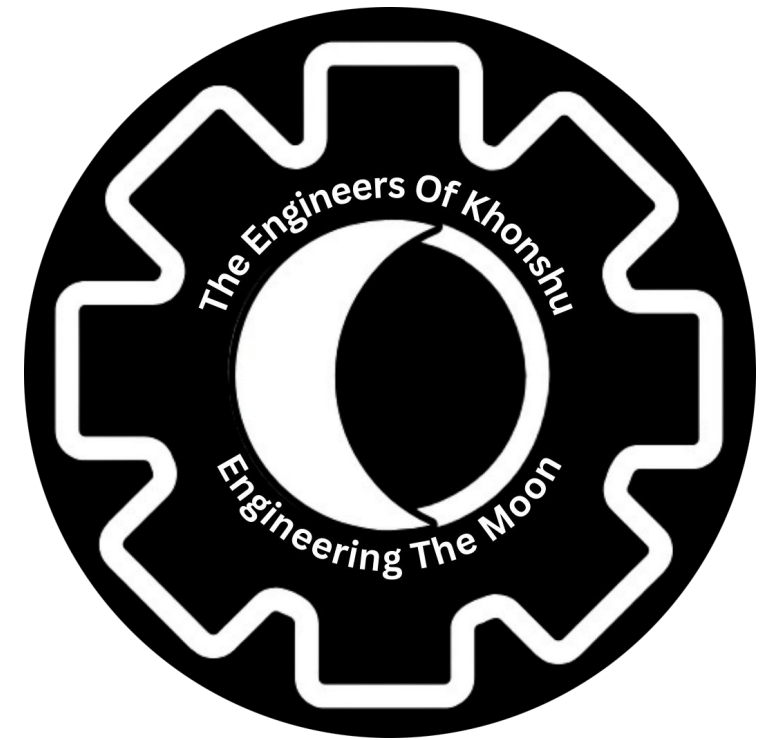
The Engineers of Khonshu

Lincoln County High School
Team 2



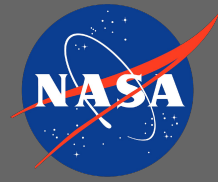
INSPIRESS

The Innovative System Project for the Increased Recruitment of Emerging STEM Students





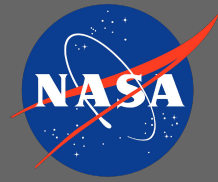
Agenda



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 - Potential Science Objectives
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 - Closing Questions
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Team



Project Manager



Tristian Hilbig

Outreach Lead



Teagan Holt

Design Lead



James Stephens

Chief Engineer



Blake Parker

Outreach Team



Rossana Sanchez

Design Team



Cameron Goodwin



Landyn Lindsey



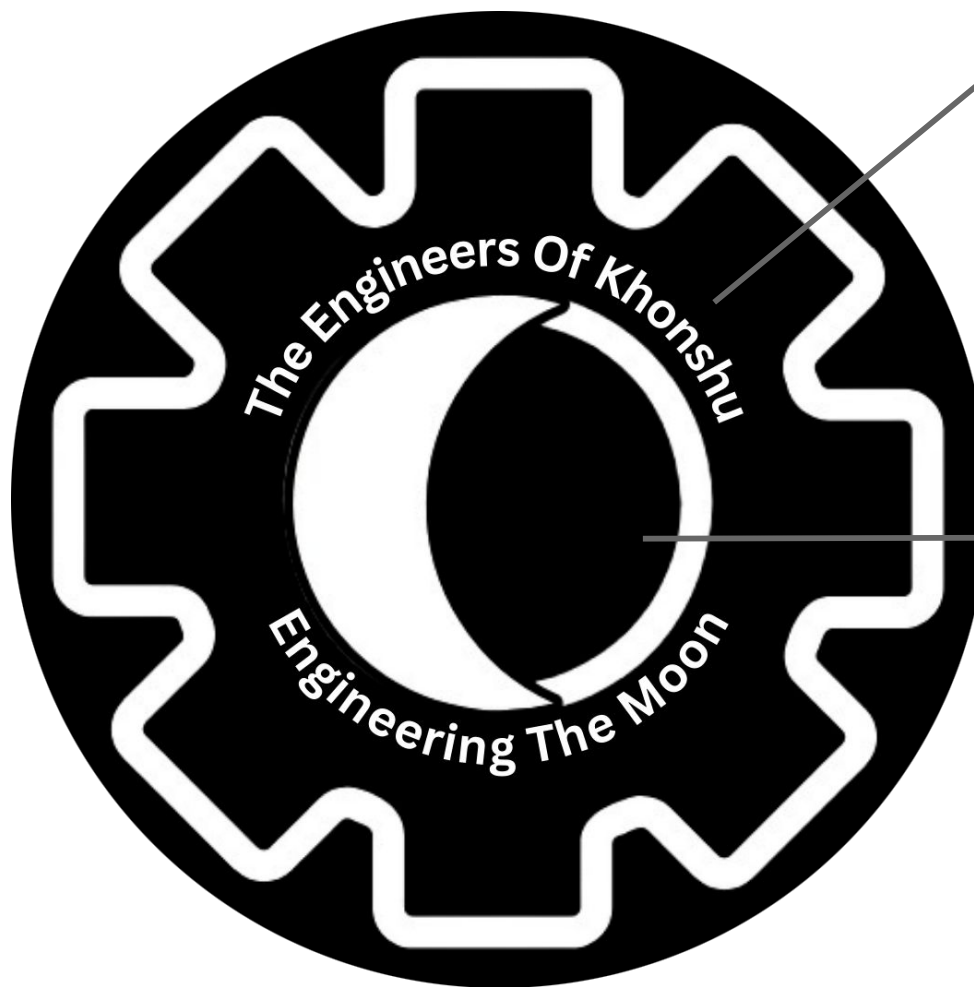
Dominick Neaves



Team Identity



The gear is representative of engineering because machines use them



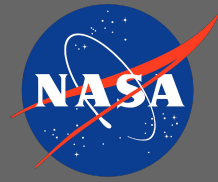
Khonshu is the Egyptian god of the Moon.

The Moon is where we are going for our mission

"Engineering the Moon"



UAH Baseline Mission - Furthering Observation and Research of Cratered Environments (FORCE) Mission



- **Orbiter (Vader)**

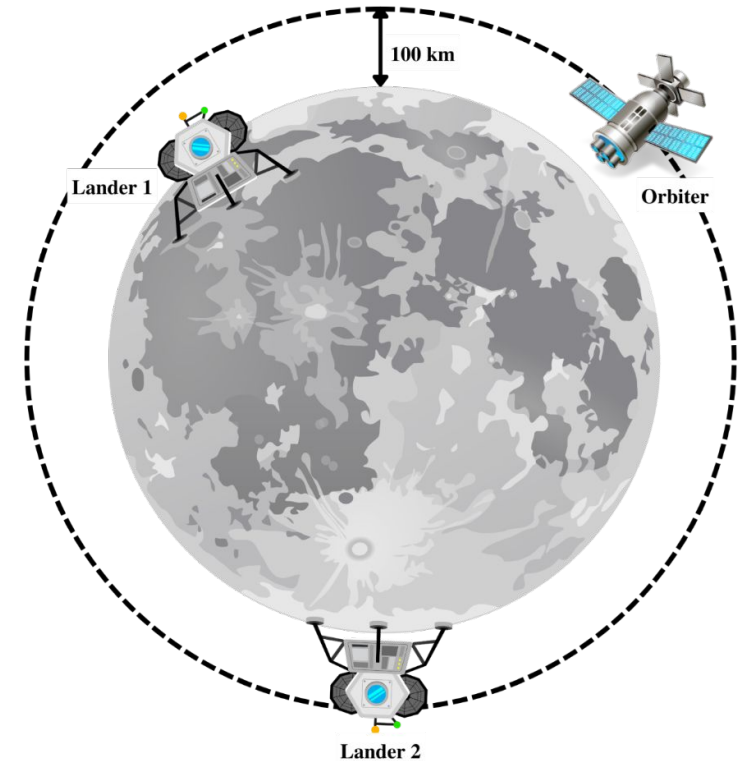
- 100 km circular polar orbit
- Two Earth-years duration (24 months)
- Communications relay to Earth

- **Lander 1 (Luke)**

- Lands at Aristarchus crater in the Aristarchus plateau located in the northern hemisphere
- 1 Earth year duration (12 months)

- **Lander 2 (Leia)**

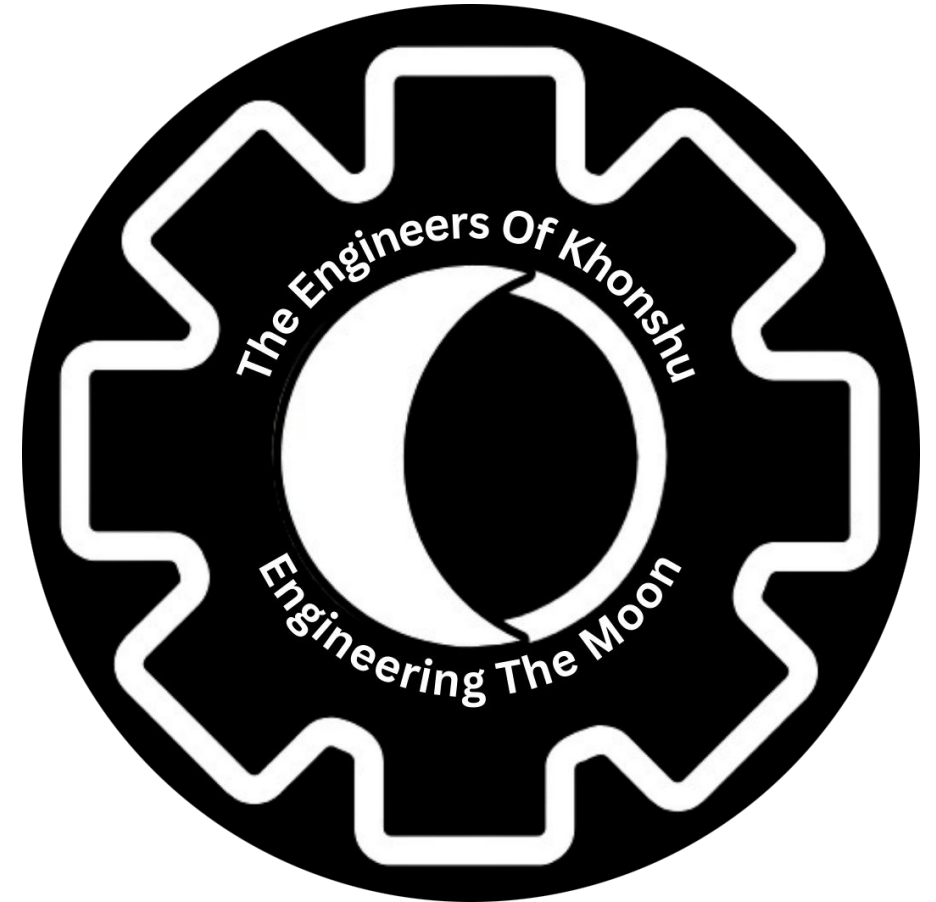
- Lands at Shackleton crater in the Aitken-Basin located in the southern hemisphere
- 1 Earth year duration (12 months)



Mission Timeline	Year 1											
	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Lunar Orbiter												
Lunar Lander 1												
Lunar Lander 2												

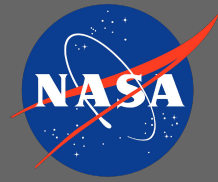
Mission Timeline	Year 2											
	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Lunar Orbiter												
Lunar Lander 1												
Lunar Lander 2												

Selection of **Science Objective**

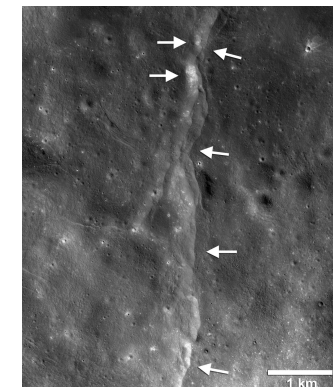
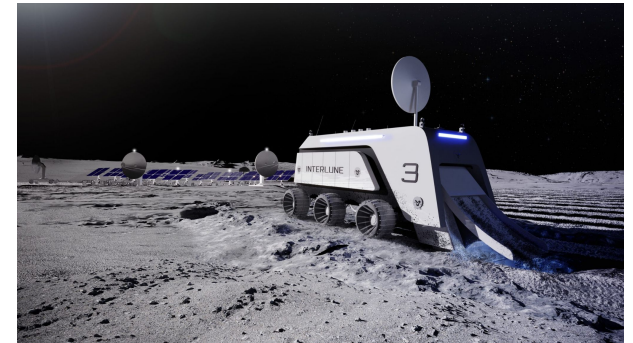




Potential Science Objectives

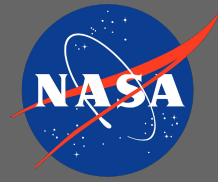


- Science Objective 1 - Can we make bricks on the Moon?
 - Explanation: Sintering lunar regolith into usable building block for further colonization of the Moon.
 - Importance: Could aid the future construction of usable habitats on the lunar surface.
- Science Objective 2 - Can we extract Helium 3 from the Moon?
 - Explanation: Theoretical Helium 3 extraction from lunar surface regolith.
 - Importance: Can aid in earth based fusion reactors.
- Science Objective 3 - What causes moonquakes?
 - Explanation: Seismology, temperature observation, and water detection via ground-penetrating radar.
 - Importance: Would help with understanding of moonquakes, average temperatures, and possible water sources on the Moon.





Science Objective Trade Study



Figures of Merit (FOMs)	Weight	Can we make bricks on the Moon?		Can we extract Helium 3 from the Moon?		What causes Moonquakes?	
		Raw Score	Weighted	Raw Score	Weighted	Raw Score	Weighted
Interest of Team	9	9	81	3	27	3	27
Applicability to other science fields (broadness)	1	9	9	1	1	1	1
Mission Enhancement	1	1	1	1	1	1	1
Measurement Method (easy to obtain)	9	9	81	3	27	9	81
Understood by the Public	9	9	81	3	27	9	81
Creates excitement in the public ("wow factor")	3	3	9	1	3	1	3
Ramification of the answer	3	3	9	3	9	1	3
Justifiability (nice, neat package), (self-consistent)	1	1	1	1	1	1	1
TOTAL			272		96		198



Science Objective Chosen

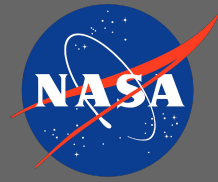


- Science Objective #1 - "Can we make bricks on the Moon?" won
 - Explanation: This machine will show whether bricks can be made from the lunar soil. It will also include other instruments to measure things that may affect the bricks.
 - Importance: Lunar habitats could be built on the Moon. This would make it easier for scientists to perform experiments on the Moon.





Science Traceability Matrix



Science Objective	Measurement Objective	Measurement Requirement	Instrument Selected
Can we make bricks out of the Lunar regolith?	Sintering Lunar Regolith	-Reaches 1200 Degrees	Thermocouple
	Measure how radiation affects the bricks	-Can measure the radiation level of the brick.	Geiger Counter
	Measure how much regolith is required for a brick	-Can measure weight	Weight Scale
	Position and orientation of payload	-Can report the rover's position and orientation	Industrial Measurement Unit (IMU)



Instrument Requirements

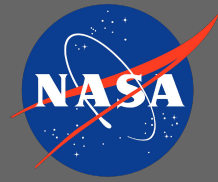


Instrument	Mass (kg)	Power (W)	Data Rate (Mbps)	Dimensions (cm)	Lifetime	Frequency	Duration
Thermocouple	0.05	N/A	1.0×10^{-4}	Wire (Length TBD)	7 Days	1 Hours	5 Seconds
Geiger Counter	0.032	0.7	1.0×10^{-6}	5.3x3.2x1.4	7 Days	2 Hours	5 Minutes
Weight Scale	0.258	5	1.0×10^{-6}	25.8x26.6 x4.3	7 Days	20 Minutes	5 Seconds
IMU	0.003	2	5.18	2x2.7x.4	7 Days	Constant	Constant

Thermocouple: Rosemount™ 214C RTD Temperature Sensor **Geiger Counter:** piDOSE-DCD/FM



Support Equipment Requirements



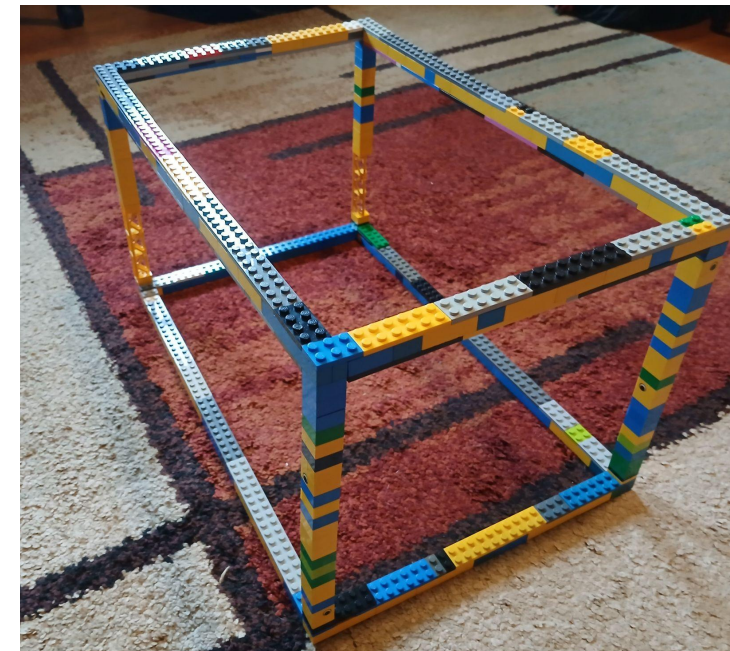
Component	Mass (kg)	Power (W)	Data Rate	Dimensions (cm)	Other Technical Specifications
Raspberry Pi Compute Module 5	0.01	2.3	64 GB Onboard Storage	5.5 x 4 x .47	64-bit processor running at 2.4 GHz 4kp60 HEVC video decoder
Arduino Uno Rev3	0.025	0.25	Flash Memory 32 KB SRAM 2 KB EEPROM 1 KB	6.86 x 5.34	14 I/O pins
ISISPACE GNSS Active Patch Antenna	0.018	0.0475	150 Mbps	7 x 7 x .97	Antenna peak gain 5.2 (1575 MHz) Frequency Range: 1572 - 1578
Batteries	1.25	N/A	12.8 V	19.5 x 33.6 x .14	9 Batteries composed of 72 different cells providing nominal 12.8 V at 27,000 mAh



Payload Design Requirements

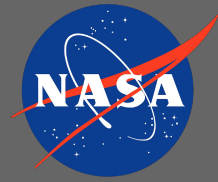


- Project requirements
 - Weighs up to 10 kg
 - Fits in a 44 cm x 24 cm x 28 cm space when stored.
 - Can perform objective away from spacecraft
 - Causes no damage to the main spacecraft
- Functional requirements
 1. Deploy
 2. Take Measurements
 3. Collect Data
 4. Provide Power
 5. Send Data
 6. House Payload

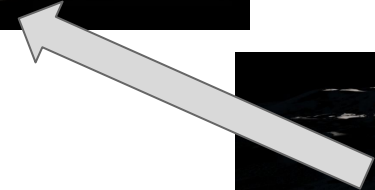
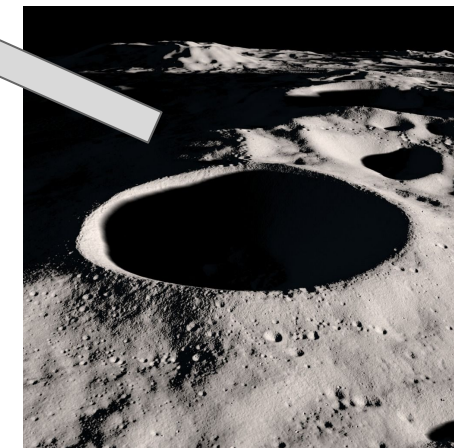




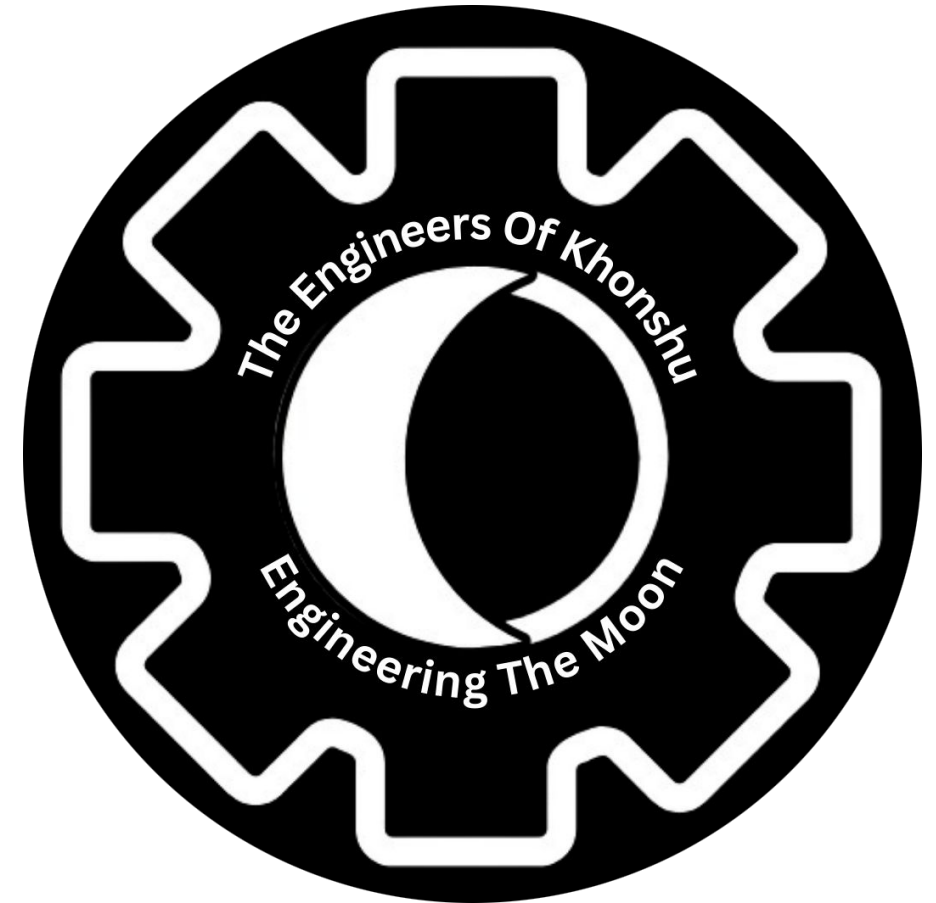
Payload Destination



- Payload Destination
 - Our payload will be delivered to the Shackleton Crater by Lander 2. This crater is located on the Moon's south pole.
- Environmental Conditions
 - Crater floor $\sim -248^{\circ}\text{C}$
 - Average temperature $\sim -183^{\circ}\text{C}$
 - Radiation ~ 1.36 millisieverts
 - Pressure $\sim 3 \times 10^{-15}$ atm
 - Gravity $\sim 1.62 \text{ m/s}^2$

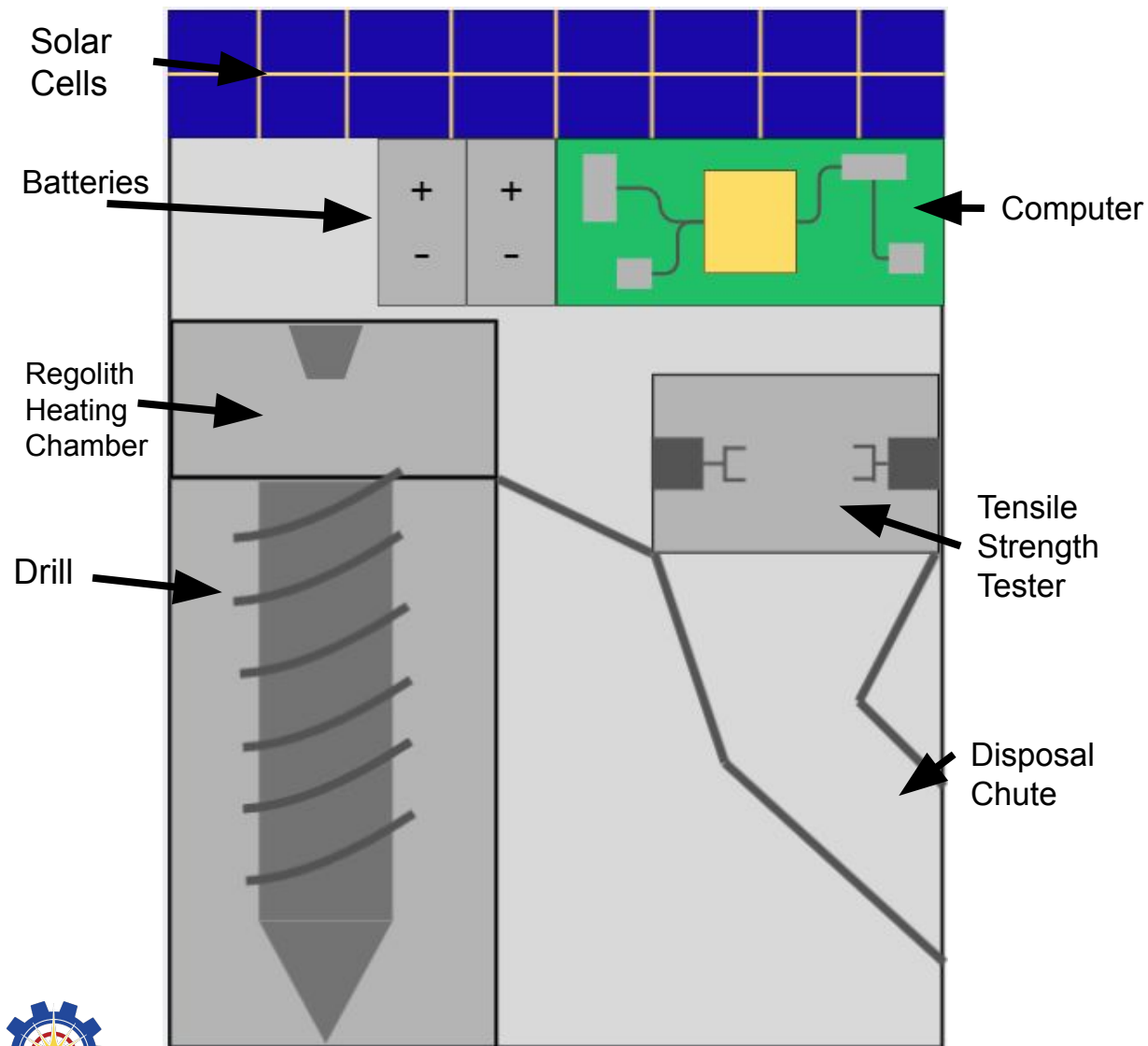


Selection of
Payload Concepts





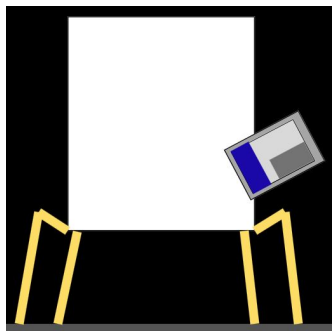
Concept 1: Ra



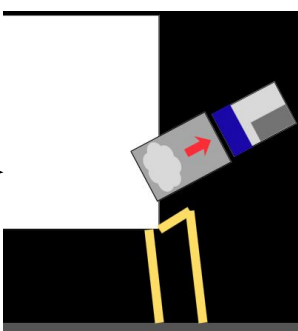
- Solar Cells - Collect solar energy to recharge batteries
- Batteries - Provide power to machine
- Computer - Controls function of machine and sends back acquired data
- Drill - Drills into ground and collects regolith
- Heating Chamber - Melts the regolith into a brick
- Tensile Strength Tester - determines the tensile strength of the bricks by testing how much force is needed to break them
- Disposal Chute - Disposes of bricks after they have been tested



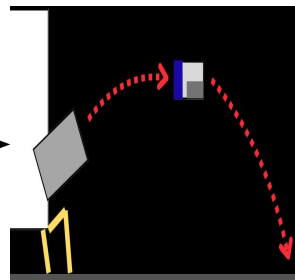
Ra - Concept of Operations



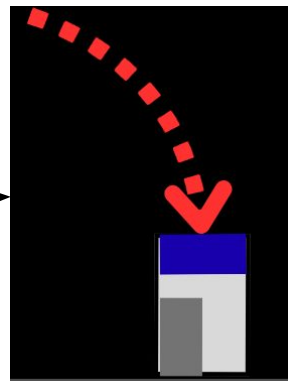
1. Initial Conditions:
Carried to Shackleton crater by Lander 2. Deployed shortly after lander makes touchdown on moon.



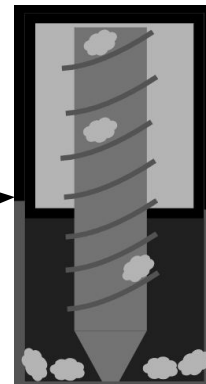
2. Deployment:
Deployed at Shackleton crater by Lander 2. Duration = ~ 1 second.



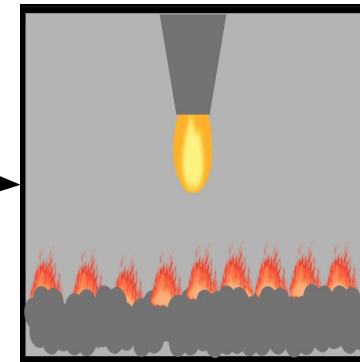
3. Trajectory:
Has an arching trajectory away from Lander 2. Duration = ~15 second.



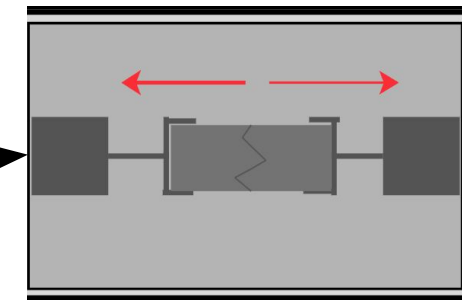
4. Lands after deployment



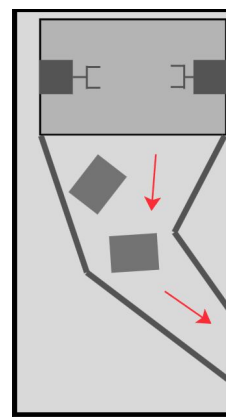
5. Collects Regolith from current position



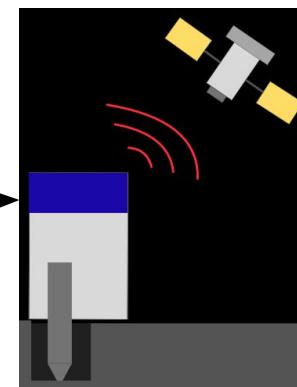
6. Melts the regolith into bricks



7. Acquires tensile strength of brick by pulling it apart.



8. Tested Brick is disposed of

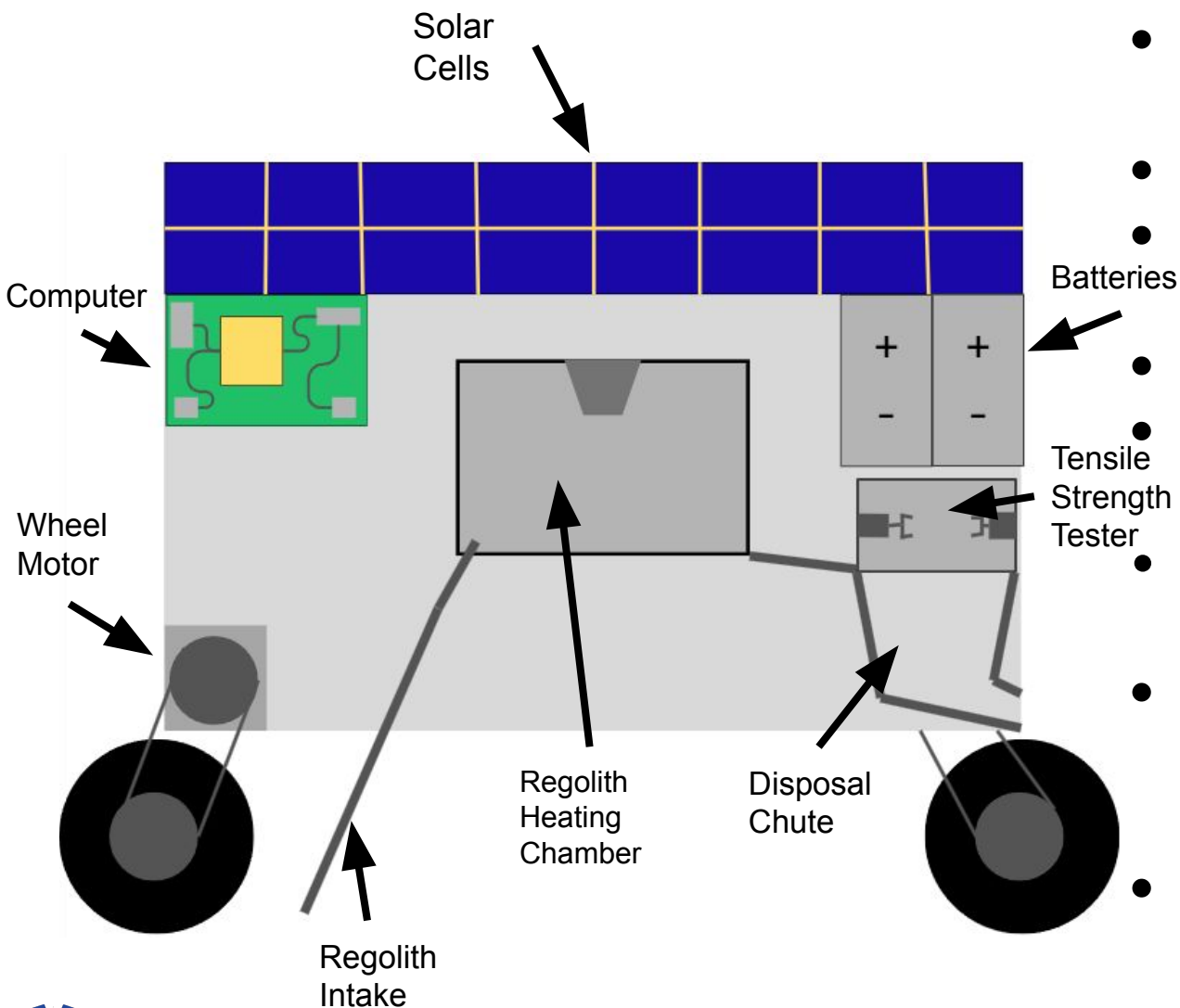


9. Ending Conditions: Once regolith runs out, report findings to orbiter.

Mission Duration = ~2 Weeks



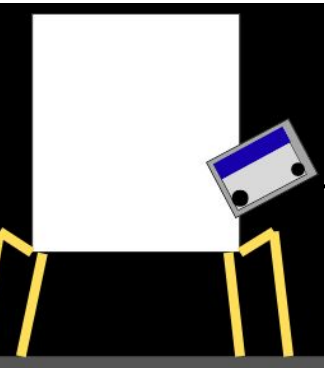
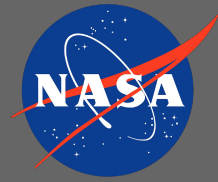
Concept 2: Geb(keb)



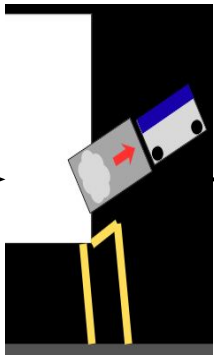
- Solar Cells - Collect solar energy to recharge batteries
- Batteries - Provide power to machine
- Computer - Controls function of machine and sends back acquired data
- Wheel Motor - Drives vehicle forward
- Regolith Intake - Picks up loose regolith from ground
- Regolith Heating Chamber - Melts the regolith into a brick
- Tensile Strength Tester - determines the tensile strength of the bricks by testing how much force is needed to break them
- Disposal - Disposes of brick after it has been tested



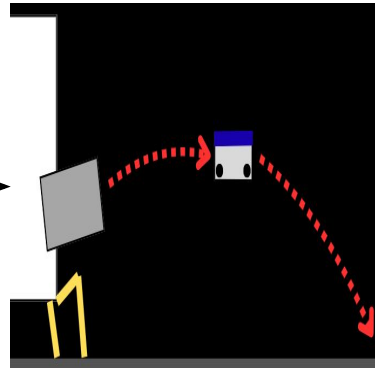
Geb - Concept of Operations



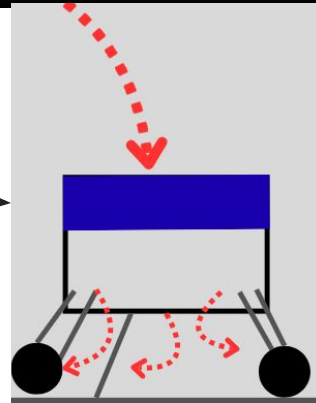
1. Initial Conditions:
Carried to Shackleton crater by Lander 2. Deployed shortly after lander makes touchdown on moon.



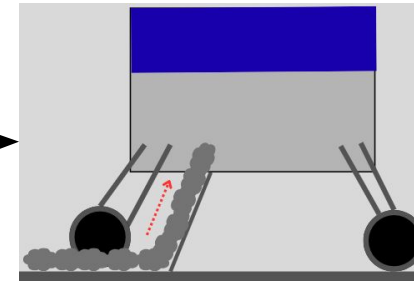
2. Deployment:
Deployed at Shackleton crater by Lander 2. Duration = ~ 1 second.



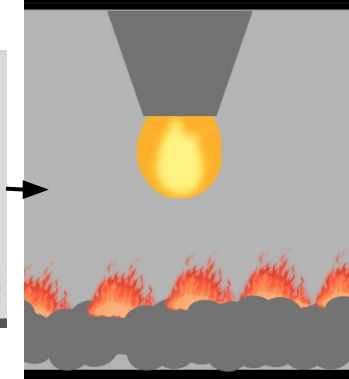
3. Trajectory:
Trajectory is away from Lander 2. Duration = ~ 15 second.



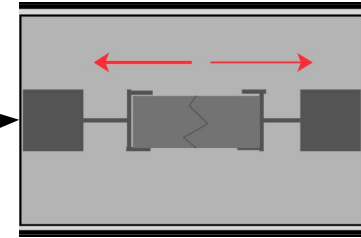
4. Lands after deployment. Wheels Fold out and Regolith intake ramp drops down



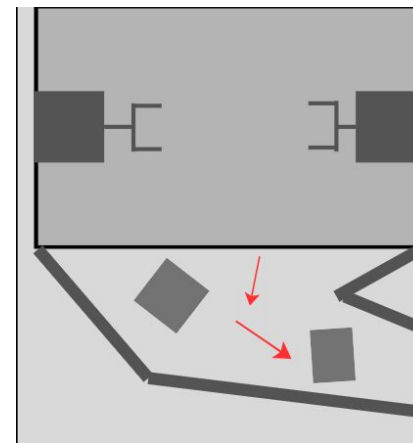
5. Drives around and collects loose regolith



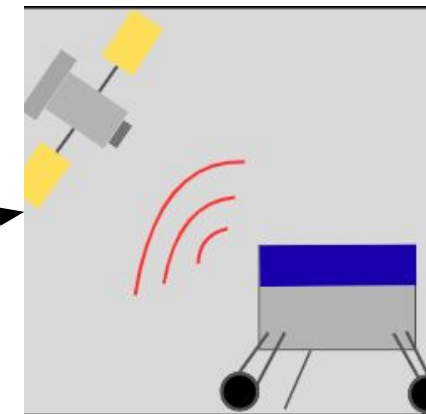
6. Melts the regolith into bricks



7. Acquires tensile strength of brick by pulling it apart.



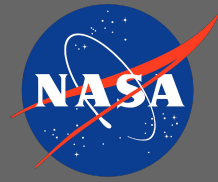
8. Tested Brick is disposed of



9. Ending Conditions: After each test, reports findings to orbiter. Minimum Duration = ~ 1 week Possible Duration = ~ 2 Years



Concept Selection Trade Study



Figures of Merit (FOMs)	Weight*	RA		GEB(KEB)	
		Raw Score	Weighted	Raw Score	Weighted
Science Objective	9	3	27	9	81
Likelihood Project Requirement	9	3	27	9	81
Science Mass Ratio	1	1	1	3	3
Design Complexity	3	3	9	9	27
ConOps Complexity	3	9	27	9	27
Likelihood Mission Success	9	9	81	9	81
Manufacturability	3	9	27	9	27
Range*	3	1	3	3	9
Durability*	9	9	81	3	27
Stability*	3	3	9	9	27
TOTAL			292		390



Material Selection Trade Study



FOMs	Weight*	AA1050-0 Wrought Aluminum		IMI 550 Titanium		Polyether Ether Ketone (PEEK)	
		Raw Score	Weighted	Raw Score	Weighted	Raw Score	Weighted
Density	9	9	81	3	27	9	81
Cost	3	3	9	1	3	3	9
Manufacturability	3	9	27	3	9	9	27
Durability	9	1	9	3	27	9	81
TOTAL			124		66		198

Aluminum: <https://www.spacematdb.com/spacemat/datasearch.php?name=AA1050-0>

Titanium: <https://www.spacematdb.com/spacemat/datasearch.php?name=IMI%20550>

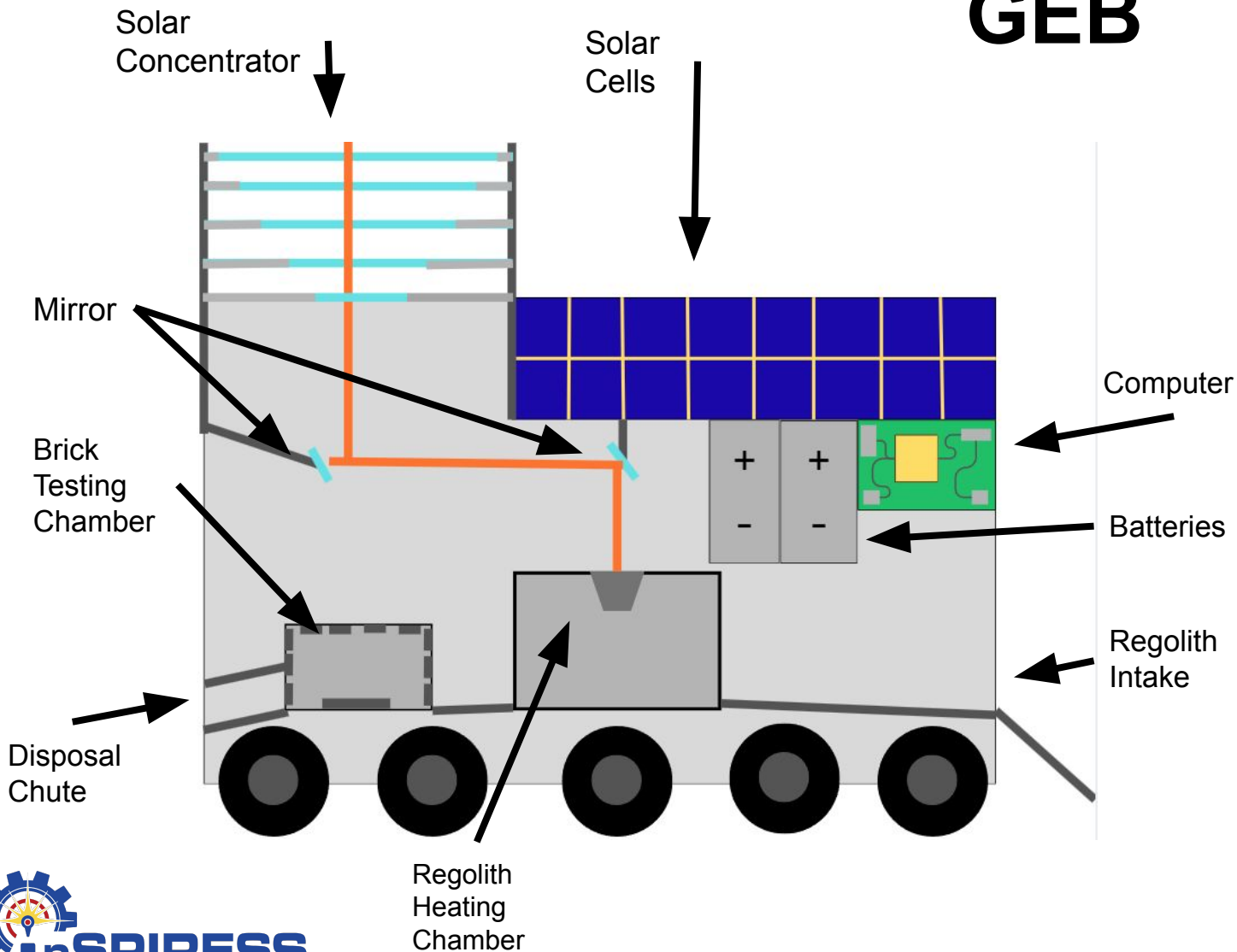
PEEK: <https://www.spacematdb.com/spacemat/datasearch.php?name=PEEK%20GF30>



Chosen Concept



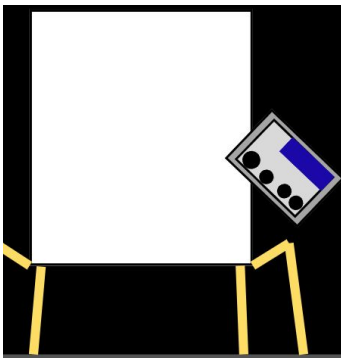
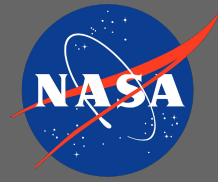
GEB



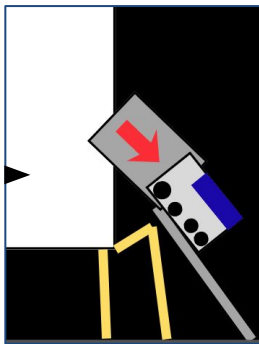
- Positives
 - Can get samples from a variety of locations
 - If more regolith is needed, then can drive to a new location
- Negatives
 - Can get stuck on lunar surface
 - External Parts can be easily damaged
- Changes Since Trade Study:
 - Wheels repositioned for better space efficiency
 - Solar concentrator attached to top



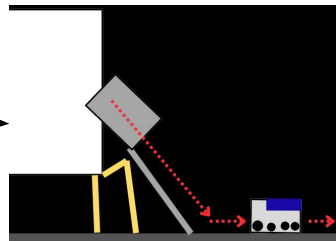
Geb - Concept of Operations



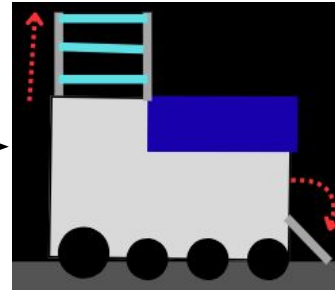
1. Initial Conditions:
Carried to Shackleton crater by Lander 2.



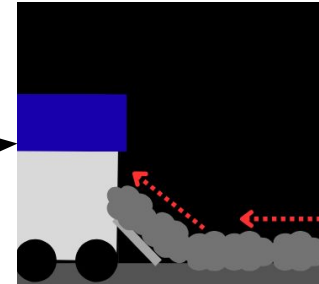
2. Deployment:
Deployed at Shackleton crater by Lander 2. Rolls out using ramp
Duration = ~ 3 second.



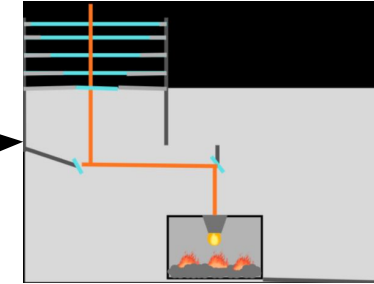
3. Trajectory:
Trajectory is parallel with the ramp deployed from Lander 2.
Duration = ~ 15 second.



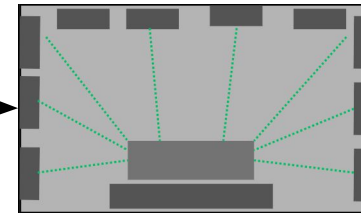
4. After touching ground, Solar concentrator and Regolith collector deploy



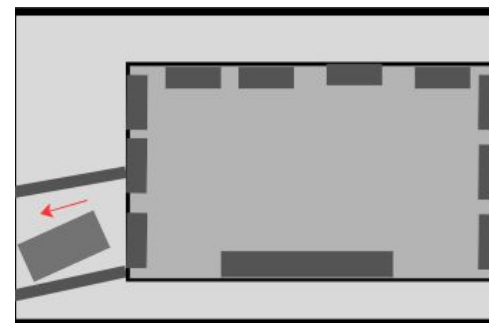
5. Drives around and collects loose regolith



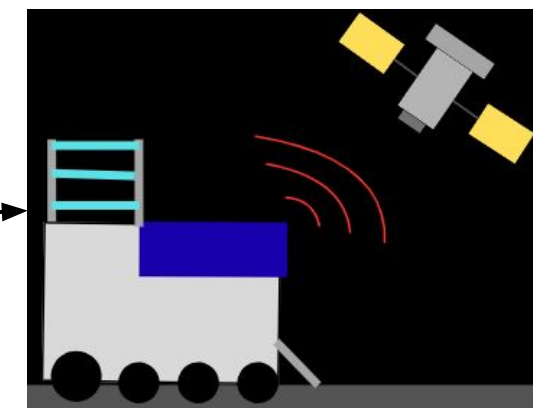
6. Compresses and heats the regolith into bricks



7. Acquires various measurements by testing the brick

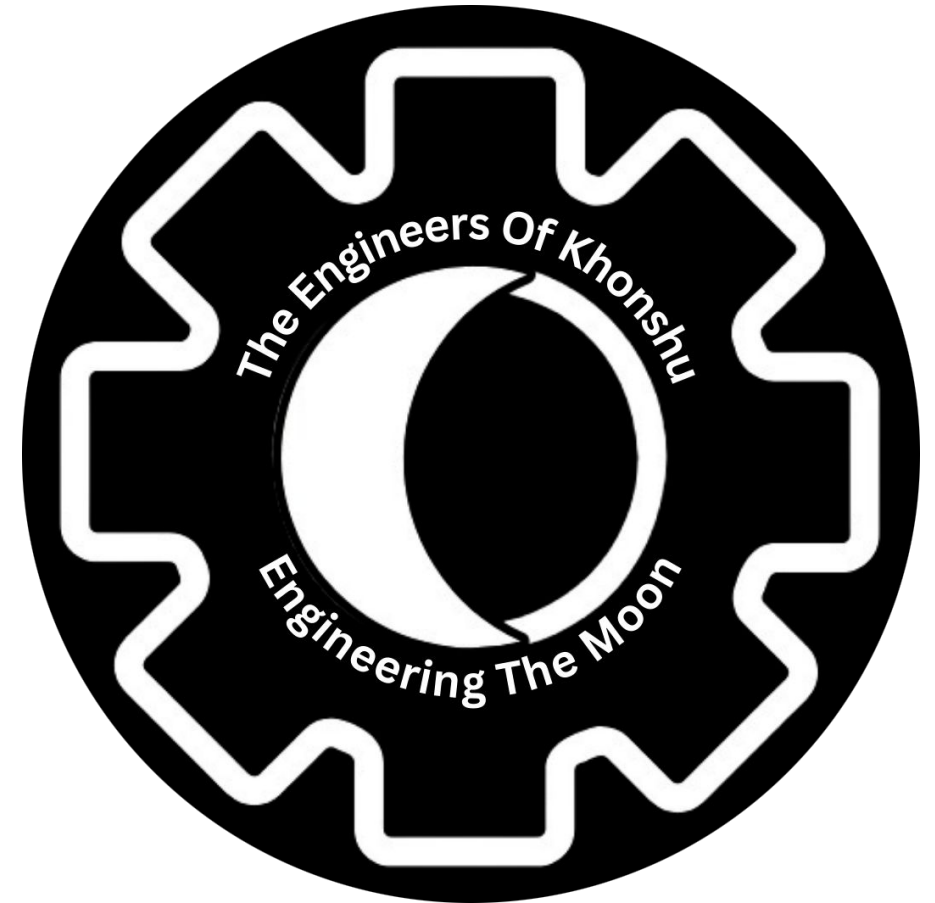


8. Tested Brick is disposed of



9. Ending Conditions: After each test, reports findings to orbiter.
Minimum Duration = ~ 1 week
Possible Duration = ~ 2 Years

Engineering Analysis

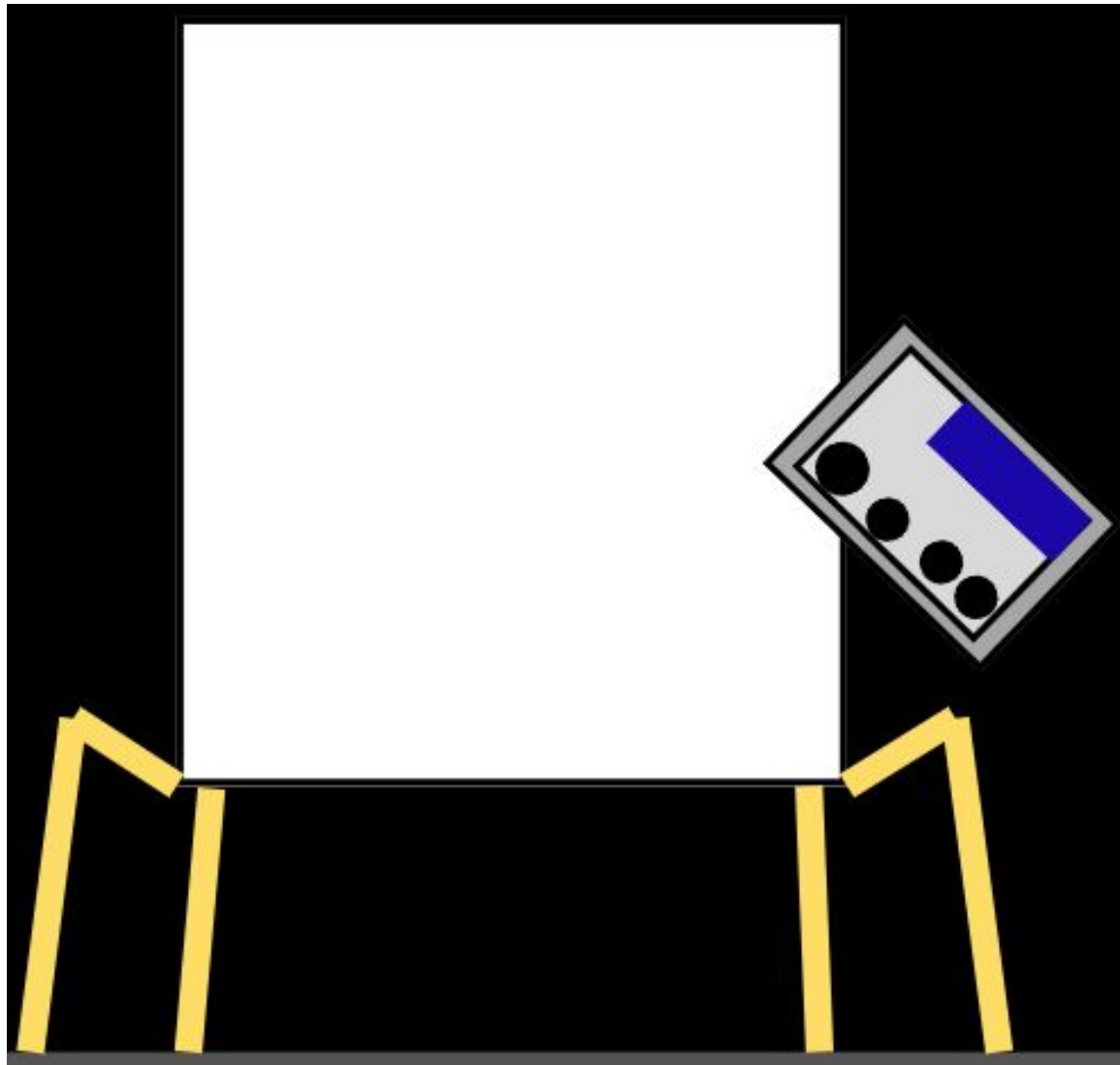




Analysis: Initial Conditions

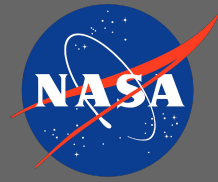


- Given:
 - At rest
 - Gravity: 1.62 m/s^2
- Assumptions:
 - Ramp Available and deployable
 - Stationary, Stable lander, angle $0^\circ \sim 12^\circ$
- Solution:
 - $v=0 \text{ m/s}$

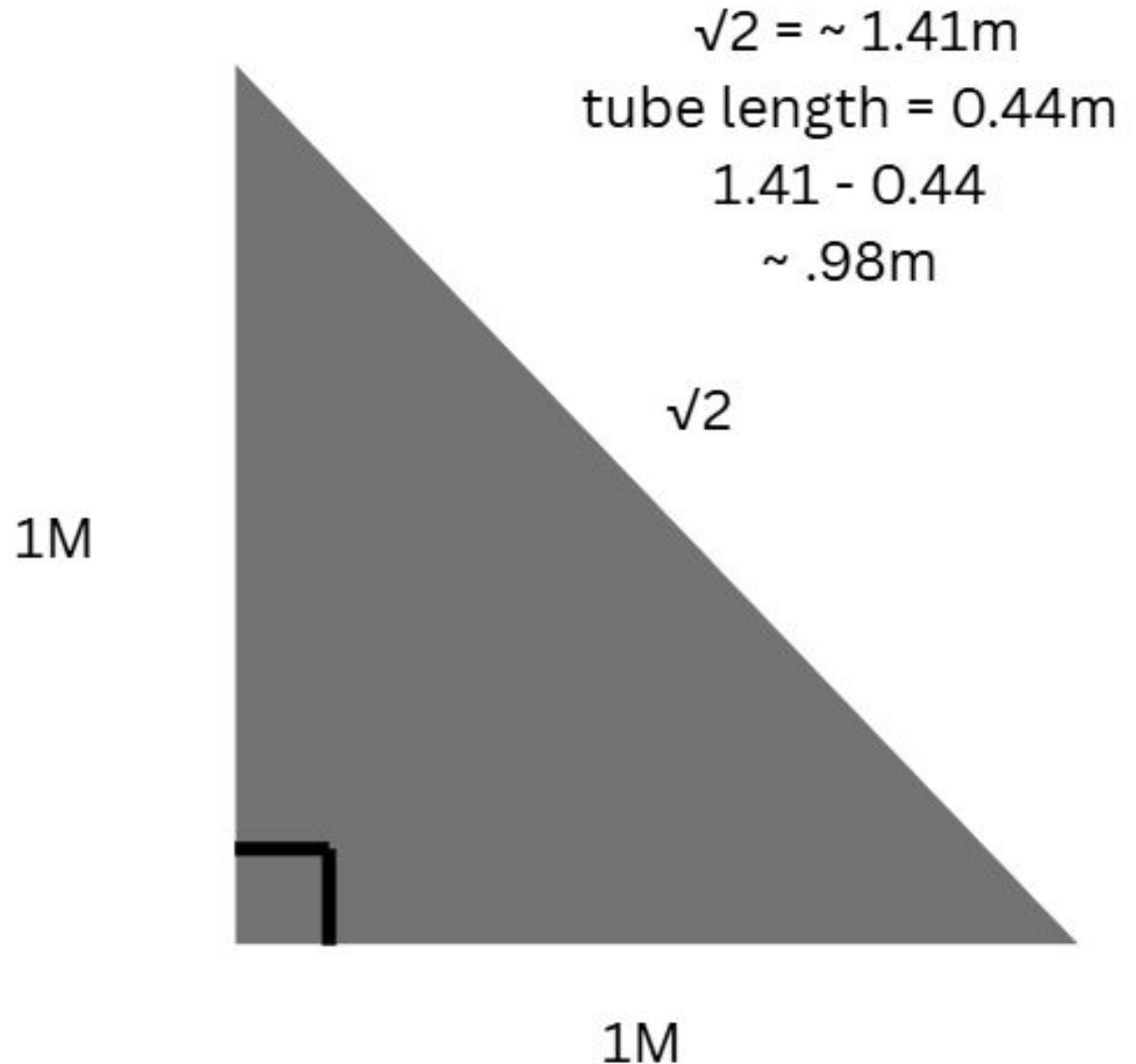




Analysis: Deployment

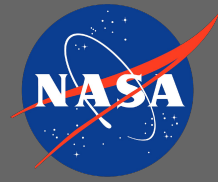


- Given:
 - Mass: 9.738 kg
 - Tube Length: 0.44m
 - $V_i = 0$ m/s
 - Starting height above ground = 1 m
 - Self propelled, so no pressure needed
- Assumptions:
 - No Friction
 - Able to go down Ramp
- Solution:
 - Ramp Angle = -45°
 - Ramp Length = ~ 0.98 m
 - Speed on ramp - 0.20 m/s
 - G-load= 0.026

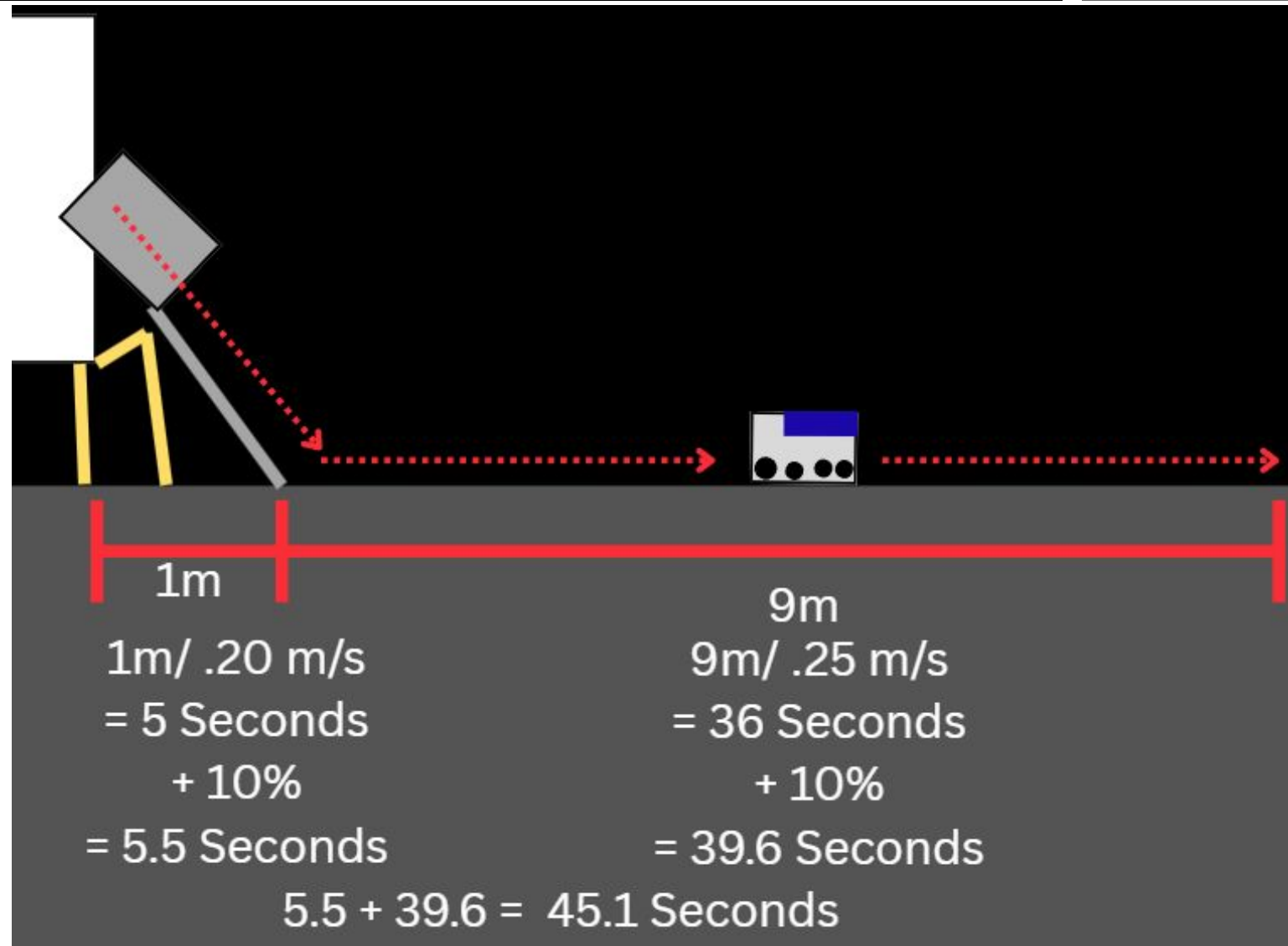




Analysis: Trajectory

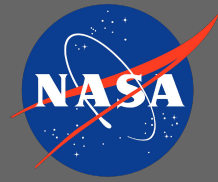


- Given:
 - $m = 9.738 \text{ kg}$
 - $g = 1.68 \text{ m/s}^2$
 - Speed on ramp = 0.20 m/s
 - Speed on flat ground = 0.25 m/s
- Assumptions:
 - No Drag
 - Start on ramp 1m Above Ground
 - 0m Starting Distance
 - Flat Terrain
- Solution:
 - Time to safe distance = 45.1 seconds

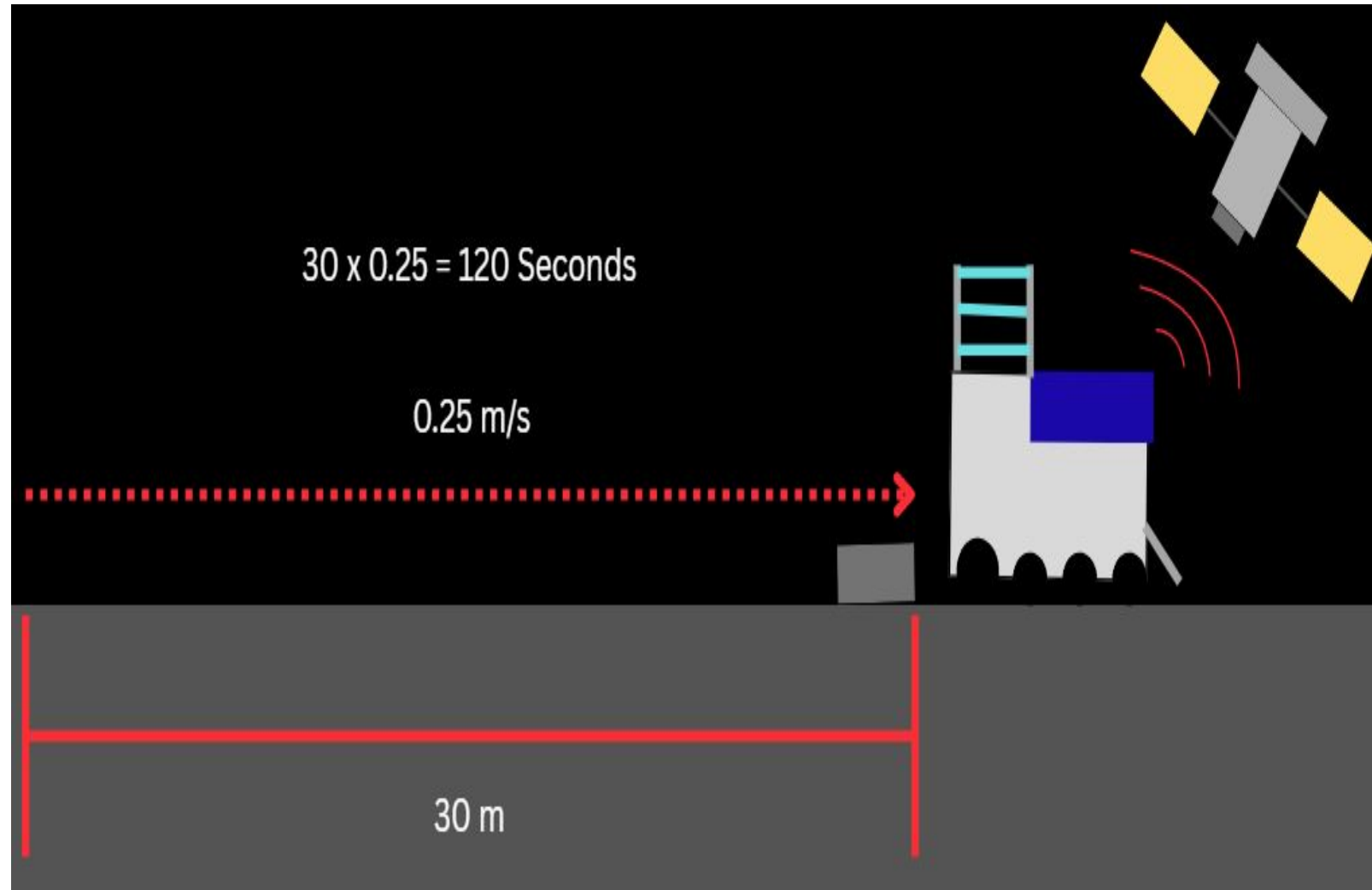




Analysis: Ending Conditions



- Given:
 - $m = 9.738 \text{ kg}$
 - $g = 1.68 \text{ m/s}^2$
 - speed = 0.25 m/s
- Assumptions:
 - No Movement Issues
 - Operates away from the Lander
 - Flat Terrain
 - No Technical or mechanical issues
- Solution:
 - Time to new spot = 2 Minutes
 - Distance to new spot = 30 m
 - $V = 0.25 \text{ m/s}$





Analysis: Batteries



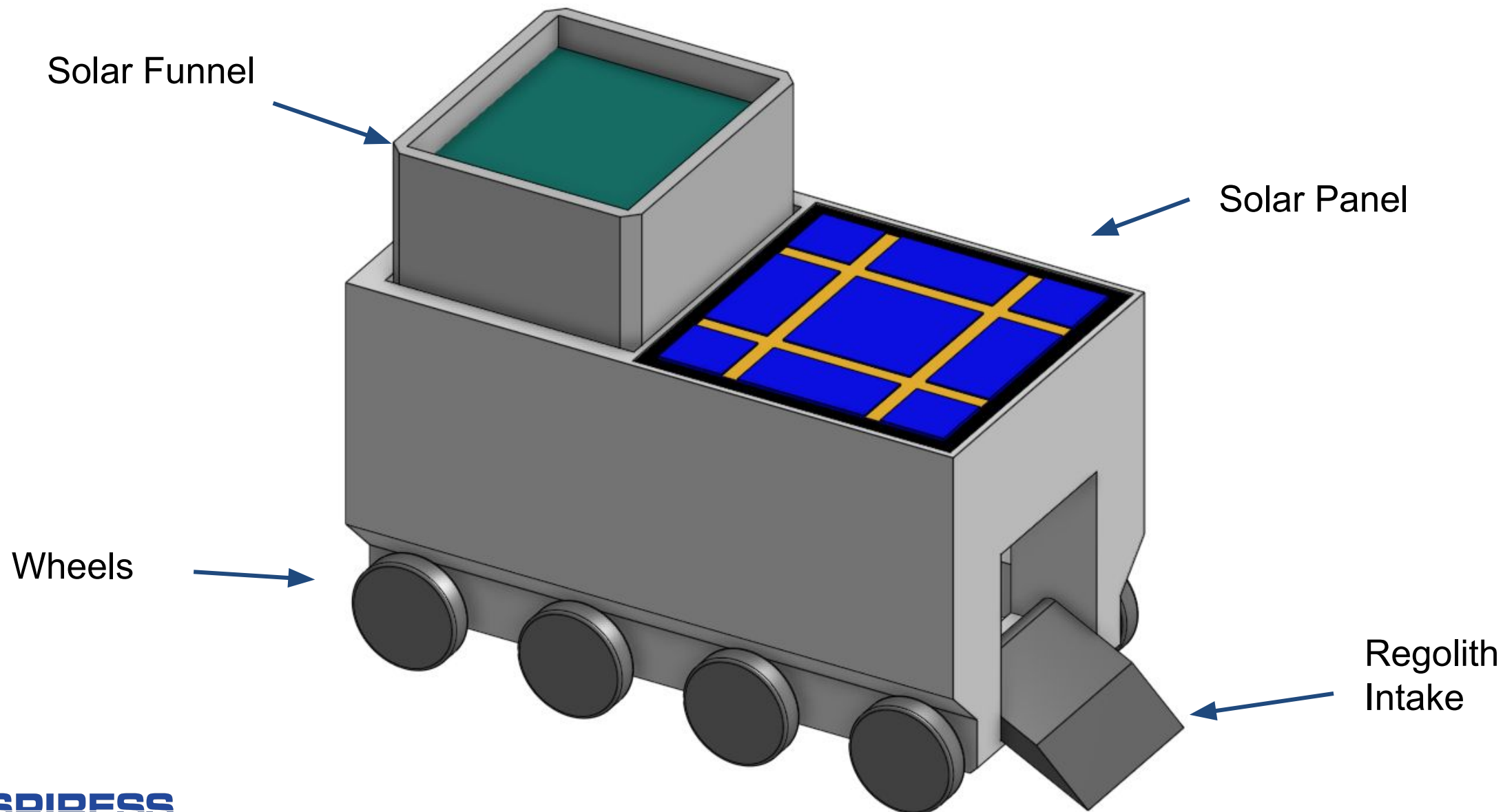
Instrument/Support Equipment	Power (W)	Operational Time (hours)	Total Power Required (W-hr)
Geiger Counter	0.7	7	4.9
Weight Scale	5	0.7	3.5
IMU	2	168	336
Raspberry Pi	2.3	168	386.4
Arduino Uno Rev 3	0.25	168	42
Patch Antenna	0.0475	168	7.98
Drive Motors x4	20	0.03	0.6
GRAND TOTAL			781.38

- Assumptions:
 - Specific mass = 400 Whr/kg
 - Sunlight available to allow solar cells to recharge batteries

Battery Mass = 1.95 kg

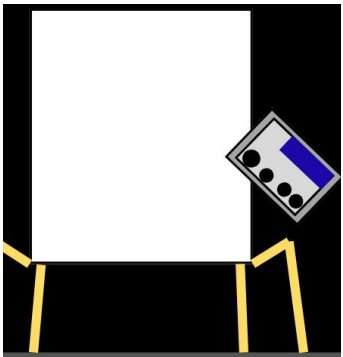
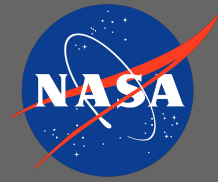


Final Design

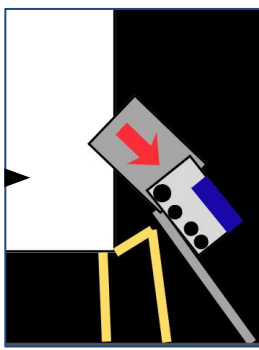




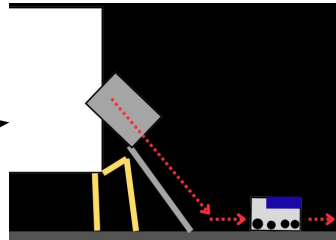
Engineers of Khonshu Mission



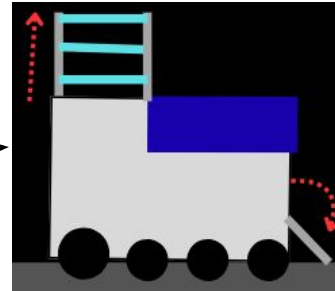
1. Initial Conditions:
Carried to Shackleton crater by Lander 2.
 $v = 0 \text{ m/s}$



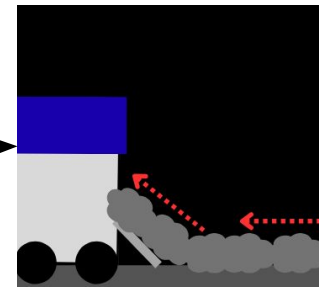
2. Deployment:
Deployed at Shackleton by Lander 2.
Uses Ramp
Ramp Length = $\sim .98\text{m}$



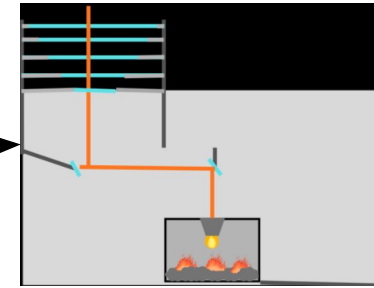
3. Trajectory:
Drives away
Drives away from lander
Time to safe distance = 45.1 Seconds



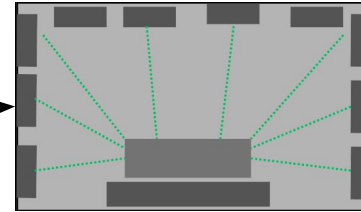
4. After trajectory, Solar concentrator and Regolith collector deploy



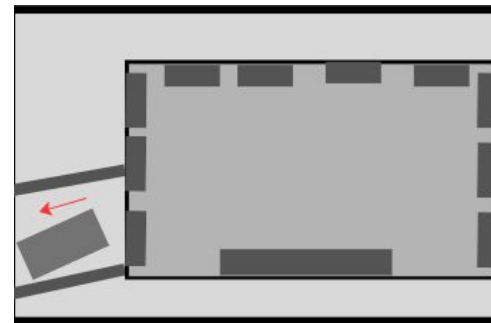
5. Drives around and collects loose regolith



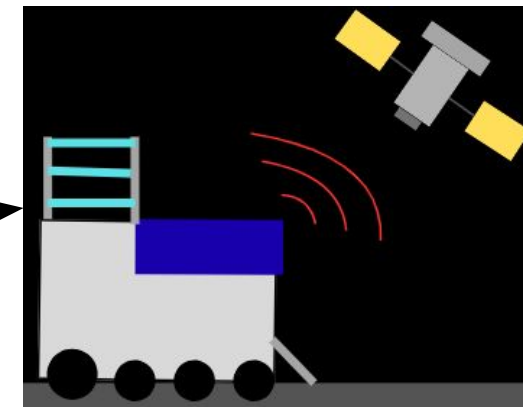
6. Uses concentrated Sunlight to melt the regolith into bricks



7. Acquires various measurements by testing the brick



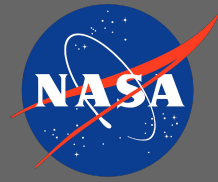
8. Tested Brick is disposed of



9. Ending Conditions: After each test, reports findings to orbiter. Then, it drives to new spot
 $V = 0.25 \text{ m/s}$
Distance = 30m
Time to New Spot = 2 Minutes



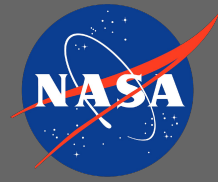
Mass Table by Function



Function	Components	Mass (kg)
Deploy	Motors, Wheels	2.24
Measure	Thermocouple, Weight Scale, Geiger Counter, IMU, Brick Melter	2.75
Collect Data	Raspberry Pi	0.01
Provide Power	Battery, Solar Cells	2.4
Send Data	Patch Antenna	0.018
House/Contain Payload	Shell, Aerogel	2.5
Totals:		9.918



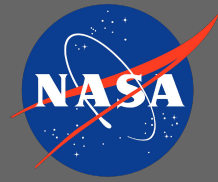
Requirements Compliance



Requirement	Verification	Compliance
• Mass ≤ 10 kg • volume $\leq 44 \times 24 \times 28$ cm	Mass: 9.648 kg Volume: 44 x 24 x 28 cm	✓
• Survive the environment ○ radiation ○ gravity ○ temperature	PEEK GF30 Radiation Resistance: 9000 Sv Specific Gravity: 1.51 Melting point 343°C Aerogel for insulation	✓
• Deploy • Measure • Collect Data • Provide Power • Send Data • House Payload	Deploy: Wheels, Motors Measure: Thermocouple, Weight Scale, Geiger Counter, IMU Collect Data: Raspberry Pi Provide Power: Battery, Solar Cells Send Data: Patch Antenna House Payload: Shell	✓



Risk Analysis

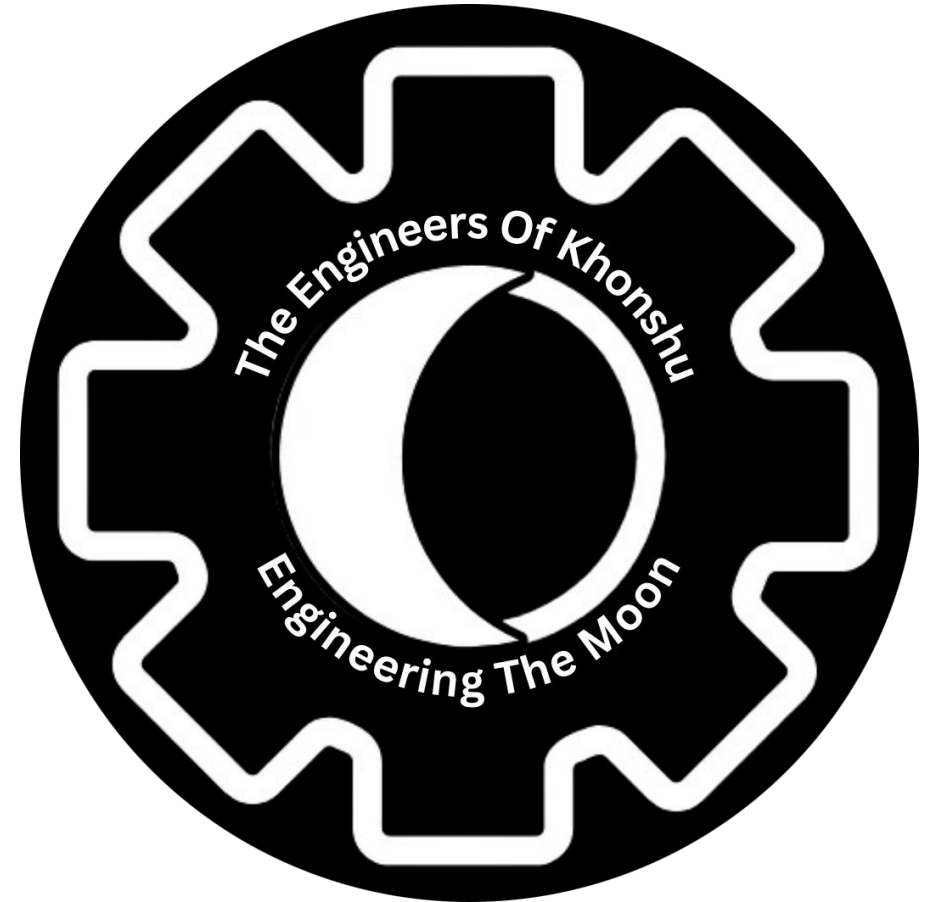


Potential risks:

- a) Loss of communication with rover
- b) Tracks on Rover get damaged
- c) Regolith jams internal working of rover
- d) Not enough regolith in the area to make a brick
- e) Regolith too loose to make bricks out of

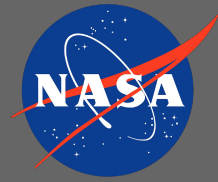
High Likelihood			
Medium Likelihood		C	
Low Likelihood		D, E	A, B
	Low Impact	Medium Impact	High Impact

Outreach Summary





ACE #1

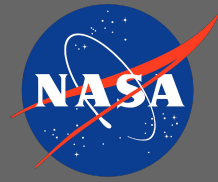


- 2/5/2026
- LCHS High School Freshman, teachers, parents, and students
- We showcased a robot previously built by robotics to show individual parts working together to perform a function

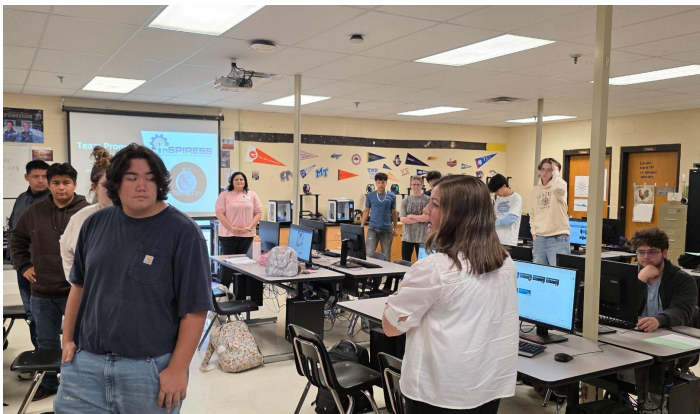




ACE #2



- 4/1/2026
- LCHS to the Kiwanis Club
- We briefly presented our slideshow to give our audience an idea of what we are doing with this project and answered any questions.
- It gives an overview of what we have done throughout the semester to complete this project





ACE #3

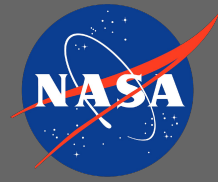


- 4/13/2026
- LCHS High School students
- We used Lego bricks to simulate the bricks we will make and use to build on the moon
- How bricks made from our payload will be used to make future buildings





ACE #4

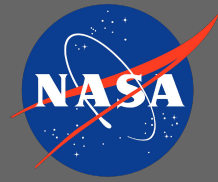


- 4/13/2026
- LCHS High School students
- We talked to students about minerals found on the moon
- The different composition of minerals affects the integrity of bricks, depending on what is present and not





ACE #5 STEAM Night



- 4/17/2026
- Over 500 people attended our community wide STEAM Night event.
- We had multiple projects going that were based around engineering and robotics to spark interest in STEAM fields.
- Brings attention and sparks interest in the engineering fields which are used throughout this project. <https://www.facebook.com/lcdoe.org>





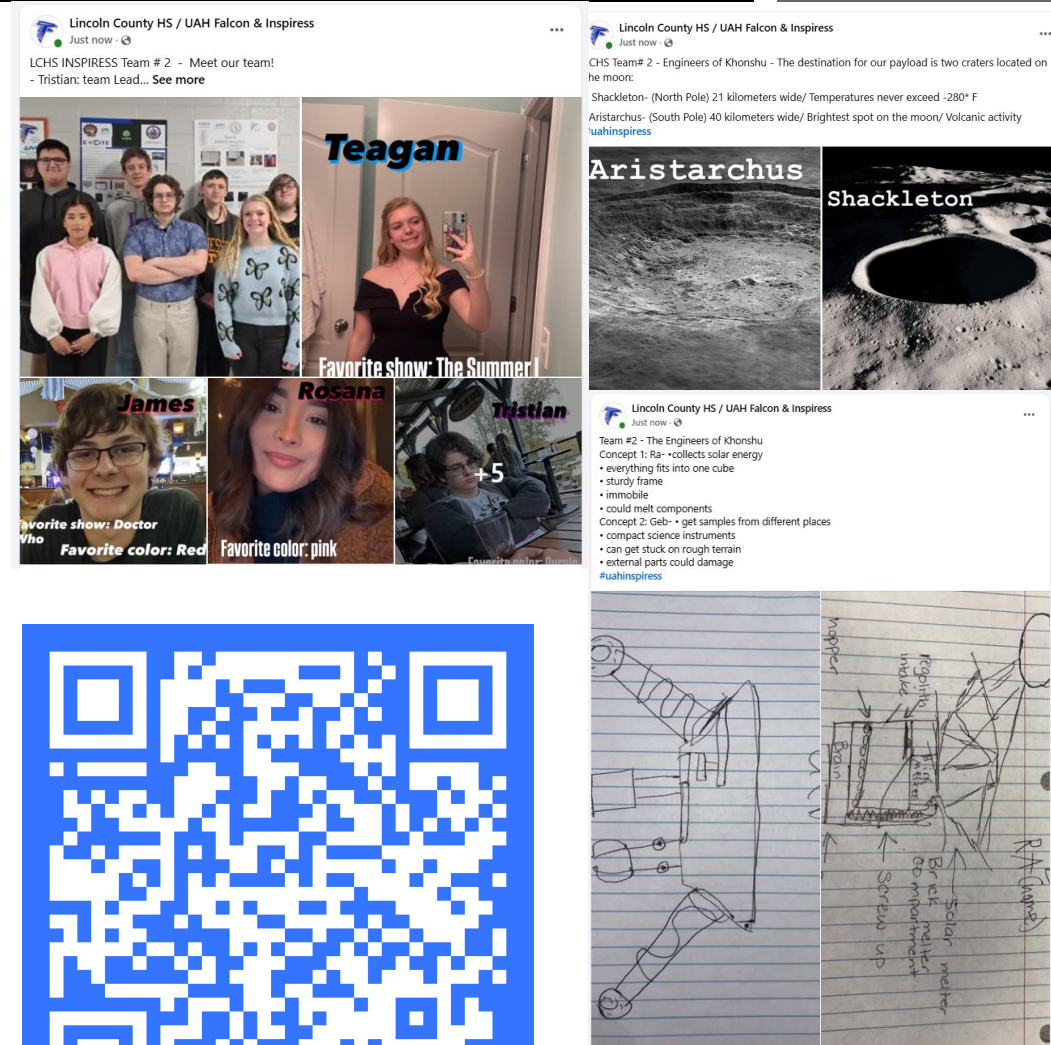
Online Outreach - Facebook



- Facebook

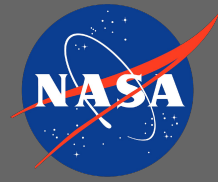
Followers- 57
Average views- 108
Total likes- 20

- Used to reach the older generations
- Post on Thursday or Fridays of each week
- Share with our friends, follow people and teams to spread our account, and post often to promote our mission
- Alternate between required posts and non-required posts each week to keep viewers interested

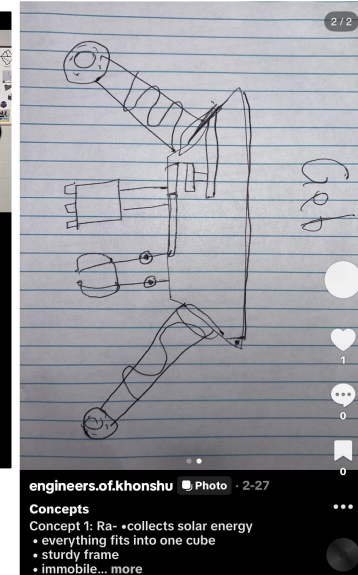
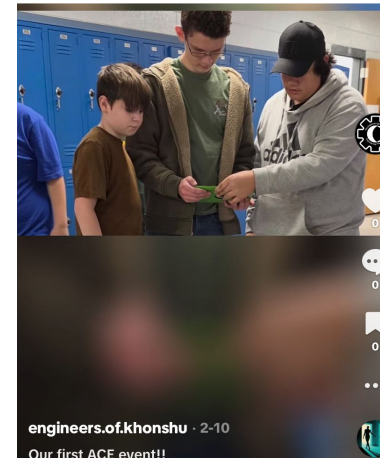




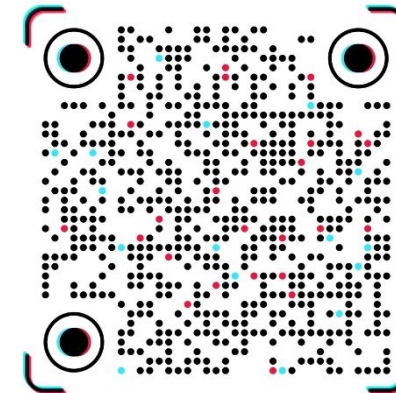
Online Outreach - Tik Tok



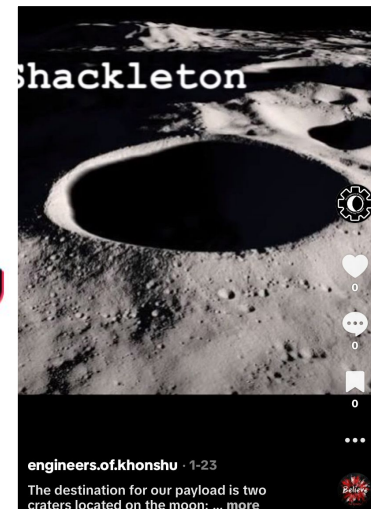
- TikTok
 - Used to reach the younger generations
 - Post on Thursday or Fridays of each week
 - Share with our friends, follow people and other teams to spread our account, and post often to promote our mission
 - Alternate between required posts and non-required posts each week to keep viewers interested
 - USERNAME: engineers.of.khonshu



Average views- 500
Total likes- 70
Followers- 5



TikTok



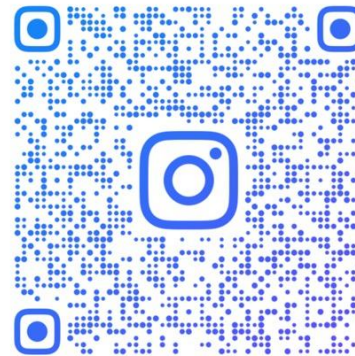


Online Outreach - Instagram

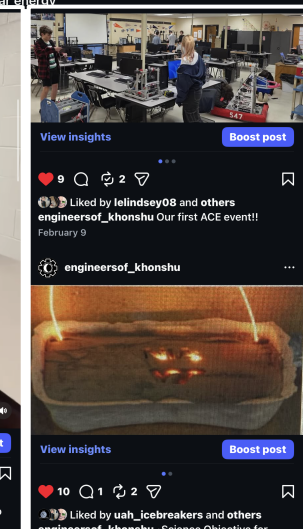
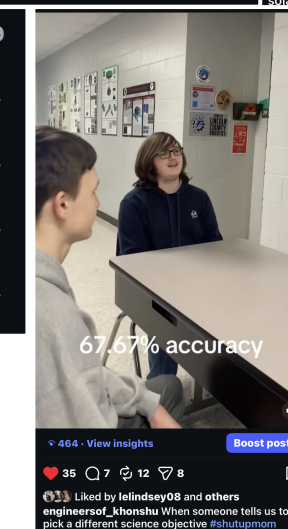
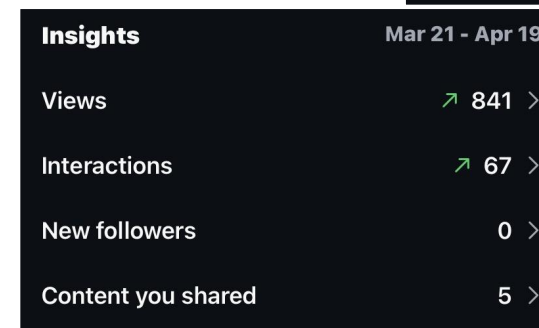
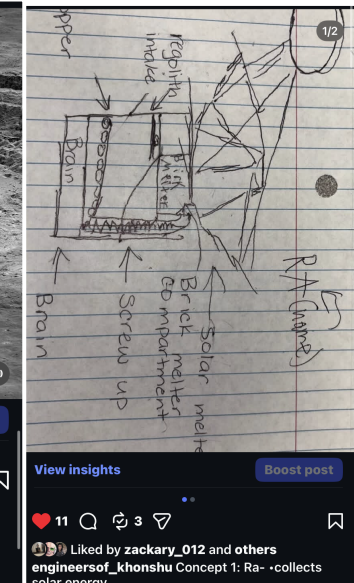
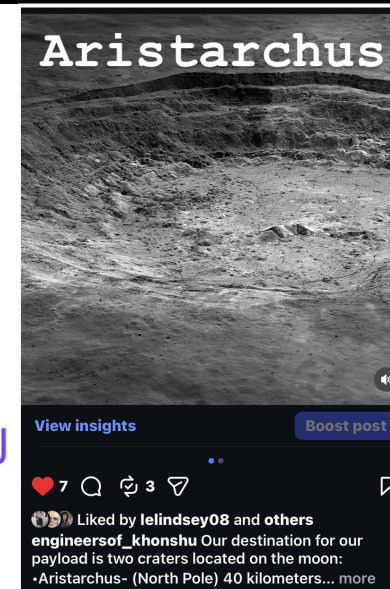


- Instagram

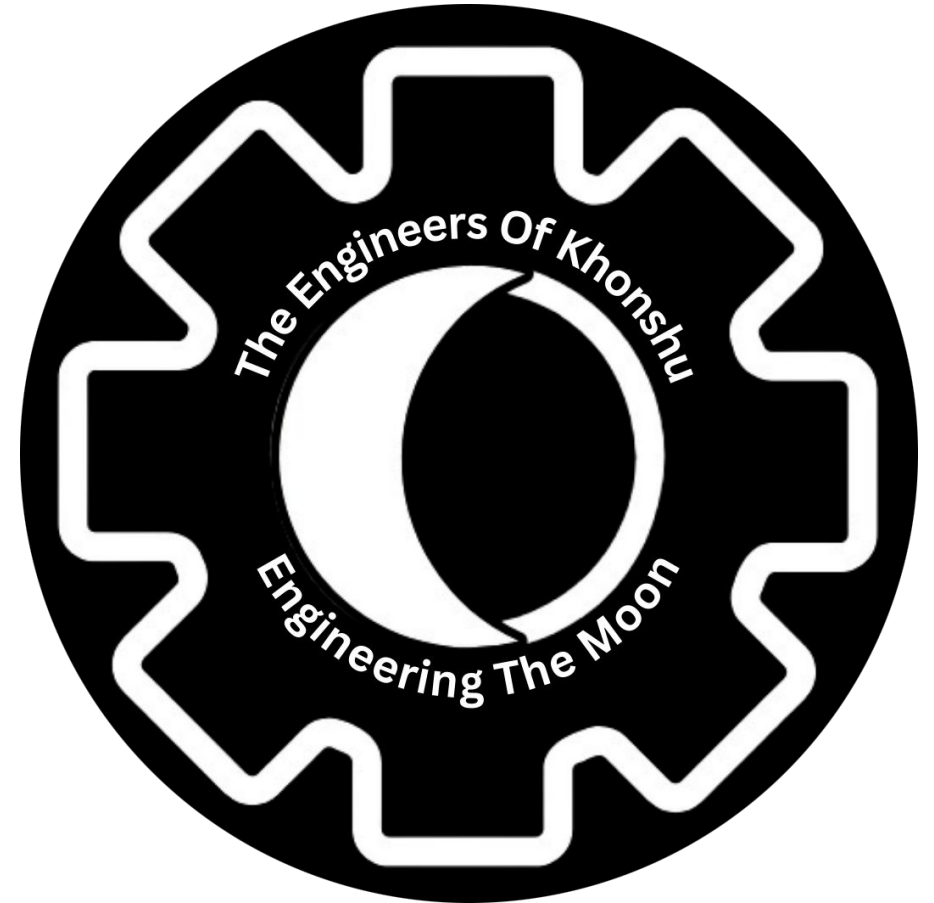
- Used to reach the younger generations
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@ENGINEERSOF_KHONSHU

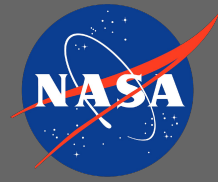


Closing Remarks





Lessons Learned



Leadership

- We learned how to manage multiple small groups performing different tasks.
- Having to manage multiple conflicting viewpoints was time consuming.
- We would try and finish a majority of the project during the first quarter when less extracurricular activities are occurring.

Design

- We learned that it is difficult to balance concept ideas with what is realistic.
- Having to fit multiple major parts inside a tiny space was more difficult than expected.
- We would figure out our design sooner, so we could have more time to make changes.

Outreach

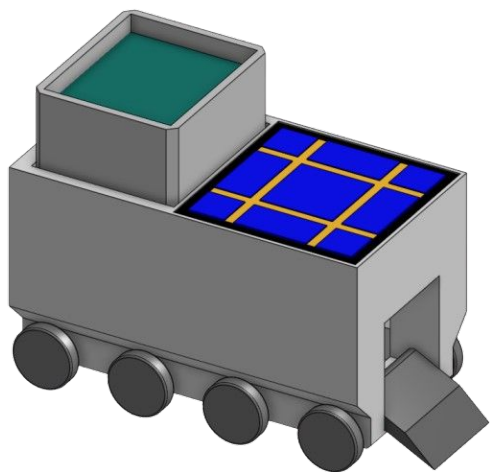
- We learned how to plan and effectively execute idea for events or post.
- Incorporating all team members into the posts and ACE events proved challenging.
- If redoing, we would plan ACE events earlier and work out the details sooner.



Summary



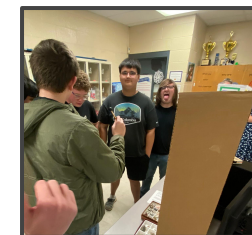
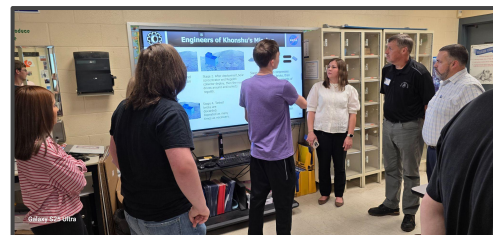
Can we make bricks on the moon?



- GEB was designed to make and test bricks using lunar regolith.
- It features a solar concentrator to use solar energy to heat the bricks
- We designed our model in the 3d CAD program Onshape

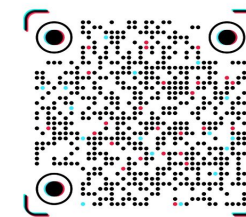
ACES

We perform hands on activities to teach about our mission and raise awareness



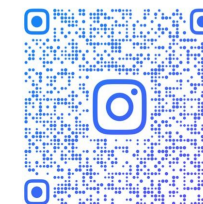
FACEBOOK

Followers- 57
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TIK TOK

Average views - 500
Total likes - 70
Followers - 5



@ENGINEERSOF_KHONSHU

INSTAGRAM

Insights		Mar 1 - Mar 30
Views	171	>
Interactions	25	>
New followers	0	>
Content you shared	1	>



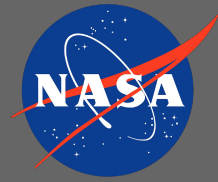
Closing Questions



Thank You For
Listening!



Sources



Slide 7:

Brick Image:

<https://www.euronews.com/next/2025/11/27/chinas-lunar-soil-bricks-return-from-space-in-major-step-toward-building-base-on-the-moon>

Helium 3 Image: <https://arstechnica.com/space/2024/03/mining-helium-3-on-the-moon-has-been-talked-about-forever-now-a-company-will-try/>

Moonquake Image: <https://www.space.com/moonquakes-moon-is-shrinking-apollo-data.html>

Slide 14:

Crater Image: <https://svs.gsfc.nasa.gov/4716/>

Moon Image: <https://en.wikipedia.org/wiki/Moon>